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DETAIL SPECIFICATION

POP-VALVE-TEST BENCH ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a pneumatic pop-valve-test bench assembly for testing air-compressor components at pressures up to 6000 pounds per square inch gage (psig).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 4940

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DEPARTMENT OF TRANSPORTATION (DOT)

Code of Federal Regulations (CFR) - Title 49

(Copies of the Code of Federal Regulations (CFR) are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

NUCLEAR REGULATORY COMMISSION (NRC)

Code of Federal Regulations (CFR) - Title 10, Parts 30 and 40.

(Copies of the Code of Federal Regulations (CFR) are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted).

(Application for copies may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Unless specified herein, materials shall be in accordance with the manufacturer's materials specifications for test bench assemblies. The materials shall be capable

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of meeting all the operational and environmental requirements specified herein. Asbestos, cadmium, and radioactive material shall not be used in this item. Radioactive material is defined by CFR, Title 10, Parts 30 and 40, and other radioactive material in which the specific activity is greater than 0.002 microcuries per gram or the activity per item equals or exceeds 0.01 microcuries. Recovered materials shall be used to the maximum extent practicable (see 4.4.1).

3.3 Design and construction. The test bench assembly shall consist of a bench, vise, water tank, pressure amplifier, gas cylinder, safety shield, gages, test blocks with adapters, and hose (see 4.4.1).

3.3.1 Bench. The bench shall consist of a top, legs, back and end stops, shelf, stringer, and drawers. The nominal dimensions of the bench, excluding the back and end stops, shall be 34 inches (in.) high, 34 in. wide, and 72 in. long. The bench shall withstand a uniformly distributed load of 1800 pounds (lb) without any visual deflection, permanent deformation, or breakage (see 4.4.3.1).

3.3.1.1 Top. The top shall have rounded corners (see 4.4.1).

3.3.1.2 Legs. The legs shall be provided with flared feet which shall not extend beyond the edge of the top. One or more holes shall be drilled or punched in the feet for attaching to the floor (see 4.4.1).

3.3.1.3 Stringer. A stringer shall be provided to stabilize the bench assembly (see 4.4.1).

3.3.1.4 Stops. The back and tapered end stops shall be provided (see 4.4.1).

3.3.1.5 Shelf. A shelf shall be provided under the top to store the nitrogen tank. Means shall be provided to ensure that the tank does not roll off from the shelf. (see 4.4.1).

3.3.1.6 Instrument turret. An instrument turret, shall be provided. The turret shall contain the pneumatic amplifier, receiver, gages, valves, and other components necessary to provide pressures for relief valve tests (see 4.4.1).

3.3.1.7 Drawers. Two drawers, minimum of 20 in. wide, 20 in. long, and 5 in. high each (inside dimensions) shall be provided. Each drawer shall withstand a uniformly distributed load of 100 lb without visual deflection, permanent deformation, or breakage (see 4.4.1).

3.4 Vise. A machinist-type vise with swivel base and 4.5 in. wide swivel jaw shall be mounted on the front of the bench, 12 in. from the manifold end of the bench. The vise shall be

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similar to figure 1. The vice jaw shall have a minimum opening of 5.5 in. The vise shall withstand a load of 2000 inch-pounds (in-lb) (see 4.4.1).

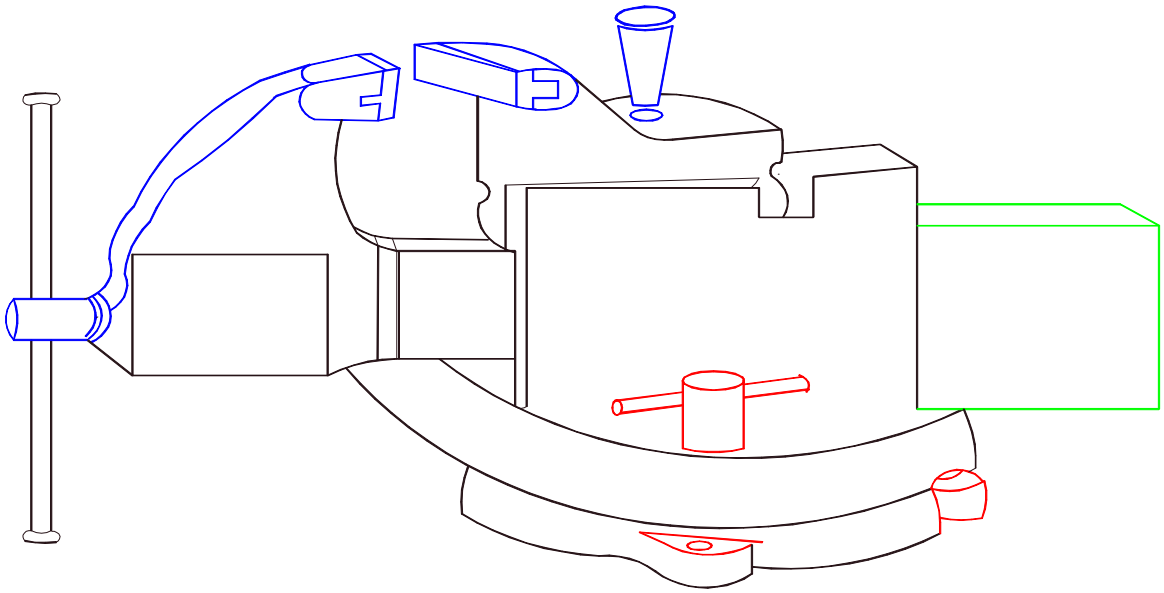


FIGURE 1. Machinist's swivel base, swivel jaw vise.

3.5 Water tank. A water tank with a minimum capacity of 15 gallons of water shall be provided. The tank shall be equipped with a drain and drain valve (see 4.4.1).

3.6 Safety shield. A safety shield shall be furnished. The shield shall be cylindrical in shape and of sufficient diameter to slide over the vise. The shield shall be mounted on a suitable stand that will permit the shield to be lowered over the vise or raised and swung out of the way when not in use (see 4.4.2).

3.7 Pneumatic system. The pneumatic system shall be arranged as shown in figure 2.

3.7.1 Pneumatic amplifier. An amplifier that requires no lubrication shall be provided. The amplifier shall include a 5 cubic inch (in^3) capacity receiver to permit precharging of the system before beginning high pressure tests (see 4.4.3.2).

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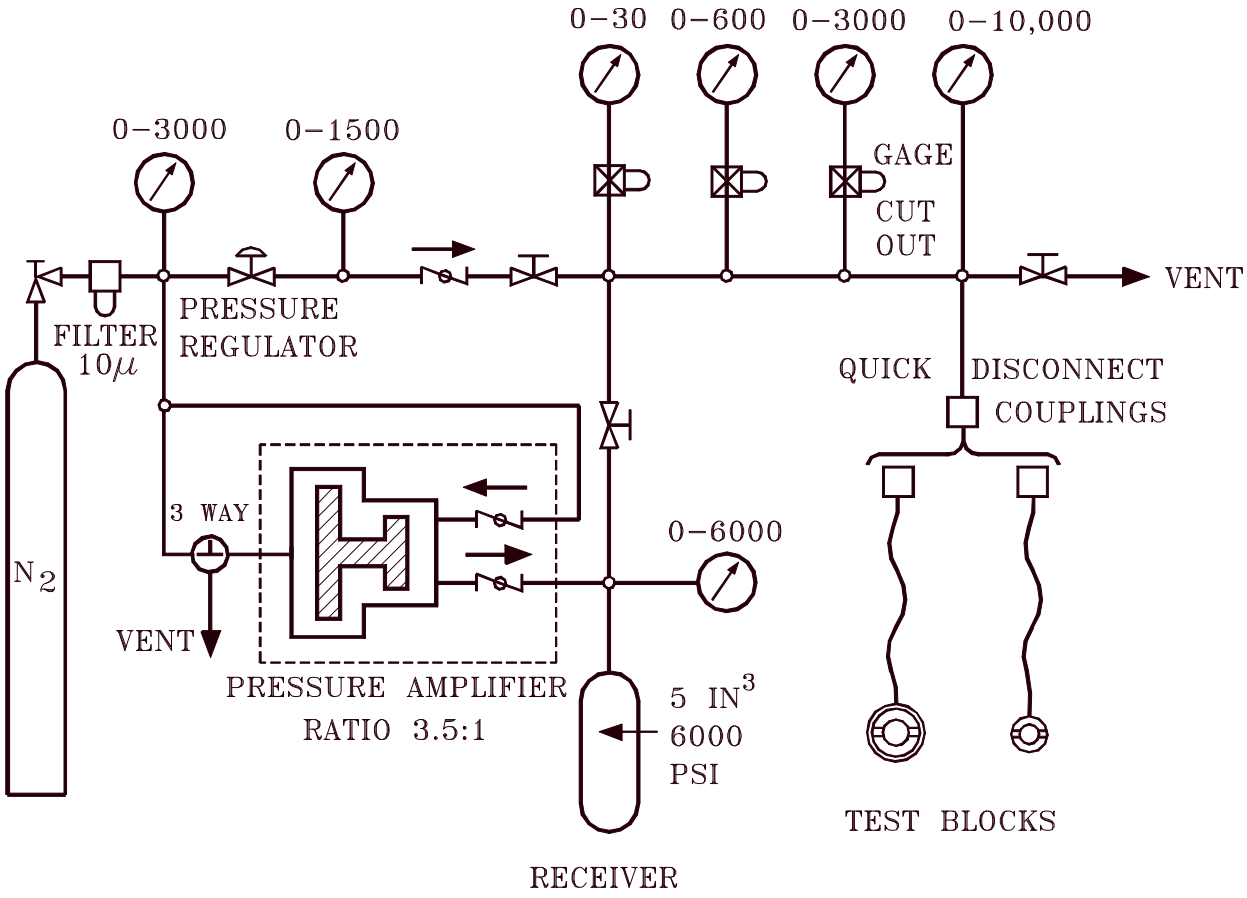


FIGURE 2. Pneumatic system.

3.7.2 Gas regulator. The gas regulator shall maintain any selected pressure within the operating range without readjustment. The regulator shall be capable of receiving pressure at 2265 psig and maintaining any selected pressure between 0 and 1200 psig with a pressure rise of not more than 1 percent of the initial set pressure in a period of 5 minutes (min). The regulator shall be equipped with two gages to indicate the source pressure and the regulated pressure. The source pressure gage shall have a scale range of 0 - 3000 pounds per square inch (psi). The regulated pressure gage shall have a scale range of 0 - 1500 psi (see 4.4.3.3).

3.7.3 Filter. A 10 micron filter with a replaceable element shall be installed in the system, between the gas cylinder and the source pressure gage (see 4.4.1).

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3.7.4 Gas cylinder. A gas cylinder shall be provided. It shall be non-shatterable and oil free. It shall have a capacity of 220 standard cubic feet of water pumped gaseous nitrogen at 2200 psig. The cylinder shall have a shut-off valve. If specified (see 6.2), cylinders shall be subject to a gunfire test as specified by the procuring activity. In addition to standard shatterproof gas cylinder markings, cylinders shall be marked with "NITROGEN (WATER PUMPED)" so as to be clearly seen from the front of the test bench assembly (see 4.4.1).

3.7.5 Gas supply line. A gas supply line shall be furnished. An adapter shall be furnished to connect the supply line to the filter. (see 4.4.1).

3.7.6 Valves. Three valves shall be installed in the system. Two shall control the high and low pressure gas into the gage manifold and one shall vent the system. (see 4.4.1).

3.7.7 Test gages. Four test gages of 0 - 30 psi, 0 - 600 psi, 0 - 3000 psi, and 0 - 10 000 psi pressure ranges shall be furnished and mounted in the turret. The gages shall be mounted on a common manifold and individually protected by a gage cutout. When the pressure exceeds the value of one gage, it shall be automatically cut out of the system, and readings indicated on the next higher gage. The gages shall be not less than 4.5 in. diameter and shall be equipped with a blow-out disc to relieve case pressure in the event of tube failure (see 4.4.1).

3.7.7.1 Gage accuracy. Gages shall be accurate to within 1.5 percent of the maximum scale reading; except that the mid-range accuracy, between 30 and 70 percent of the maximum scale reading, shall be within 0.5 to 1 percent. These requirements shall not apply below 5 percent of the maximum scale reading (see 4.4.1).

3.7.8 Hose. A hose shall be provided and connected to the pressure port of the gage manifold. The hose shall be rated at 6000 psig minimum working pressure, proof-tested to 12 000 psig with a minimum burst pressure of 24 000 psig. The hose shall be of sufficient length to reach the vise and water tank, and shall terminate with a high pressure quick-disconnect socket with a captive protective dust plug. (see 4.4.1).

3.7.9 Test blocks. Test blocks shall be furnished to accommodate valves of the following National Pipe Thread (NPT) sizes: 1/4-in., 3/8-in., 1/2-in., 3/4-in., 1-in., 1-1/4-in., 1-1/2-in., 2-in., 2-1/2-in., and 3-in. Each block shall be equipped with a high pressure quick-disconnect nipple with a captive dust plug (see 4.4.1).

3.8 Identification marking. Identification marking shall be permanent and legible and shall include, as a minimum, the manufacturer's name and part number and the National Stock Number (NSN). Special markings, if any (see 6.2), shall be as specified (see 4.4.2).

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3.9 Instruction plates. Each test bench shall be equipped with instruction plate. Plates shall be permanently attached in a location where the plates will be both legible and visible (see 4.4.2).

3.10 Treatment and painting. The portions of the test bench normally painted, shall be cleaned, treated, and painted in accordance with the manufacturer's standard practice (see 4.4.2).

3.11 Workmanship. The assembly shall be free of defects to its intended use and service life (see 4.4.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article test bench inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article test bench inspection. One first article sample (see 3.1), shall be inspected as specified in 4.2.1 and 4.2.2.

4.2.1 Examination. The first produced test bench shall be examined as specified in 4.4.2. Presence of one or more defects shall be cause for rejection.

4.2.2 Tests. The first produced test bench shall be tested as specified in 4.4.3.1 through 4.4.3.3. Failure of any test shall be cause for rejection.

4.3 Conformance inspection.

4.3.1 Examination. Each test bench shall be examined as specified in 4.4.2. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. Each test bench shall be tested as specified in 4.4.3.2 and 4.4.3.3. Failure of any test shall be cause for rejection.

4.4 Method of inspection. Bench assembly shall be inspected as specified in 4.4.1, 4.4.2, and 4.4.3.

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4.4.1 Material and design. Conformance to 3.2, 3.3, 3.3.1.1 thru 3.3.1.7, 3.5, and 3.7.3 thru 3.7.9, shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.4.2 Examinations. The test bench shall be examined for the following defects listed in table I.

TABLE I. Classification of defects.

Category	Defect	Method of examination
Major:		
101	Safety shield not as specified (see 3.6).	Visual
102	Identification marking not as specified (see 3.8).	Visual
103	Instruction plates not as specified (see 3.9).	Visual
104	Treatment and painting not as specified (see 3.10).	Visual
105	Poor workmanship (see 3.11).	Visual

4.4.3 Tests.

4.4.3.1 Static load. Conformance to 3.3.1, place a uniformly distributed load of 1800 lb on the top of the bench and 100 lb in each of the two drawers for 10 min. Any permanent deformation, cracking, or breaking shall constitute failure of this test.

4.4.3.2 Low pressure pneumatic test. Conformance to 3.7.1, adjust the regulator to a pressure of 10 psig and hold for 5 min. Then readjust the regulator to a pressure of 600 psig and hold for 5 min. Finally readjust the regulator to a pressure of 1200 psig and hold for 5 min. Leakage of the cylinder connection, filter, or regulator; or failure of the regulator to maintain the set pressure within 1 percent for 5 min. shall constitute failure of this test.

4.4.3.3 High pressure pneumatic test. Conformance to 3.7.2, operate the three-way cycling valve and precharge the amplifier receiver to 6000 psig. Attach the small test block to the test connection and plug the test block. Slowly bleed the receiver pressure through the gage manifold to the test block, observing the operation of the gage cutouts. Failure of the amplifier to increase the source pressure of 6000 psi, malfunction of the gage cutouts, leakage or inability of any component to withstand the applied pressure shall constitute failure of this test.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The bench assembly covered by this specification is intended for use in testing valves, hoses, gages, and other compressor components. Because this item is used to test some military components which have no commercial counterparts, it is military unique.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.3).
c. If first article is required (see 3.1).
d. Gas cylinder gun fire test, if required (see 3.7.4).
e. Any special marking, if required (see 3.8).
f. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Air compressor
Compressor components
Gages
Hoses
Pneumatic

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Review Activities:
Air Force - 82
DLA - CC

Preparing Activity:
Army - AT

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-45859D	**2. DOCUMENT DATE *(YYMMDD)*** 980831

3. DOCUMENT TITLE

POP-VALVE-TEST BENCH ASSEMBLY

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)***

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	**b. ORGANIZATION**	
c. ADDRESS *(Include Zip Code)*	**d. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** ***(If applicable)***	**7. DATE SUBMITTED *(YYMMDD)***

8. PREPARING ACTIVITY

a. NAME	**b. TELEPHONE *(Include Area Code)*** **(1) Commercial** (810) 574-8745 **(2) AUTOVON** 786-8745
c. ADDRESS *(Include Zip Code)* **Commander** **U.S. Army Tank-automotive and Armaments Command** **ATTN: AMSTA-TR-E/BLUE** **Warren, MI 48397-5000**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403** **Falls Church, VA 22041-3466** **Telephone (703) 756-2340 AUTOVON 289-2340**

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